Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 8 · day per timetable	8	1 period (60 min)	The Quadratic Formula	Algebra II	Topic 2 — Quadratic Functions and Equations · Lesson 2-6	Grade 10 — 10A and 10C

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Canva (final product) · Squared paper	Lesson presentation (14 slides) · "Choose Your Route" differentiation sheet · "Jeddah Season Ticket Prices" project task (A3) · "Give it a go" classwork

Competencies	Technology Embedded	Connection to student's real life
1- Finding the discriminant and using it to state the number and type of solutions of a quadratic equation. 2- Solving quadratic equations with the quadratic formula, including nonreal solutions.	1- Quizizz live diagnostic with instant gap analysis. 2- Calculator check: confirm a real solution by substitution and round only at the last step.	1- A model concert ticket price: which prices reach a revenue target, and when a target is impossible. 2- Deciding whether a target can be reached before solving anything.

Values	Soft Skills	Hard Skills
1- Accuracy — every sign is written down before it is used. 2- Precision — exact answers first, decimals only at the end.	1- Collaboration in assigned group roles. 2- Explaining in words what the discriminant tells you.	1- Using $\Delta = b^2 - 4ac$ to classify solutions. 2- Solving with $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Analyze the discriminant to identify the number and type of solutions for a quadratic equation. 2- Solve quadratic equations using the quadratic formula, including those with nonreal solutions. *Vocabulary: Discriminant ; quadratic formula.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: $b^2 - 4ac$ with negatives, $\frac{4+\sqrt{20}}{2}$, $\sqrt{-36}$. Results read as a gap map; students routed to the re-teach table, straight to the formula, or the Investigate prompt. FIKR Phase 2 — Targeted Instruction. Whole class: the quadratic formula; the discriminant; nonreal solutions; the discriminant with an unknown. Quick check on whiteboards. FIKR Phase 3 — Practice. Two identical "Give it a go" problems worked together. Students then choose a route and may switch at any time. Live feedback via Pear Deck; 2-minute micro-conferences. FIKR Phase 4 — Production. The Jeddah Season ticket price: two prices, one price and no price. AI used as critic only. Extended in the A3 project task. FIKR Phase 5 — Proof of Learning. "Time to Check" — a discriminant, a real pair and a nonreal pair. No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product including one worked non-example, and publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell on the final products, then the reflection question: "Which is harder for you — finding the discriminant, or keeping the signs right in the formula?"
Reflection for Improvement:	
Homework:	Finish the final product and upload to the LMS portfolio (any of the three formats). Practice set on Pear Deck — 12 questions, choose your route. Watch the flipped video for the next lesson, Topic 2 · Lesson 2-7 — Solving Linear-Quadratic Systems. Clinic group: bring your Time to Check paper.

Teacher name:	Mr Malek Thiab